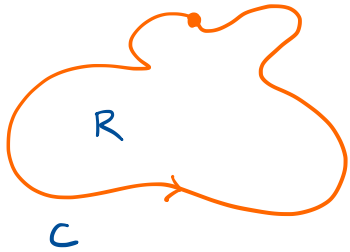
18.4 Green's Theorem

A closed curve is simple if it doesn't intersect itself. Such a curve bounds a region R. We say C is positively-oriented if R is on the left as you go around C.



Green's Theorem Suppose C is a simple closed curve that is piecewise smooth and positively-oriented. Let R be the interior of C and that $\vec{F} = \langle F_1, F_2 \rangle$ is smooth. Then

$$\oint_C F_1 \partial x + F_2 \partial y = \int_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \partial A.$$

→ Do all problems.